

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Analytical Problem Solving

Course Code:	ITA0517	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i>		
Student Name:	B.V.N.S.BHIMESWAR	Reg. No.:	192472048
Section / Batch:	CSE - AI	Date:	10-08-2026

Code of Conduct

I B.V.N.S.BHIMESWAR(192472048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.V.N.S.BHIMESWAR

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Criteria	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	Problem Understanding	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	Method Selection	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	Solution Process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	Accuracy of Calculation	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	Interpretation of Results	

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

1. GREY LEVEL TRANSFORMATION

(a) Explain the purpose of grey level transformation in image enhancement.

Problem Understanding

Grey level transformation is an image enhancement technique used to modify the intensity values of pixels in an image. It improves the visual appearance of an image and helps highlight important details.

It maps an original grey level value (**r**) to a new grey level value (**s**).

General Transformation

$$s = T(r)$$

Where:

- **r** = Original pixel intensity
- **s** = Transformed pixel intensity
- **T(r)** = Transformation function

Grey level transformation is mainly used to:

- **Improve image contrast**
- **Highlight important intensity ranges**
- **Enhance image details**
- **Make important regions easier to identify**

(b) Apply a negative transformation to the pixel values.

Method Selection

The appropriate method is **Negative Transformation**.

Formula

$$s = (L - 1) - r$$

Where:

- **L** = Number of grey levels
- **r** = Original pixel intensity
- **s** = Transformed pixel intensity

For an **8-bit image**:

$$L = 256$$

Therefore:

$$s = 255 - r$$

Solution Process

For every pixel, subtract its intensity value from **255**.

Original Pixel (r) Transformed Pixel (s = 255 - r)

0	255
50	205
100	155
150	105
200	55
255	0

Example

If **r = 100**

$$s = 255 - 100$$

$$s = 155$$

Therefore:

$$100 \rightarrow 155$$

Accuracy of Calculation

The transformed values are:

255, 205, 155, 105, 55, 0

All values are calculated using the formula:

$$s = 255 - r$$

Interpretation of Results

The **negative transformation** reverses the intensity values.

Dark pixels become bright, while bright pixels become dark.

For example:

$$0 \rightarrow 255$$

255 $\rightarrow$ 0

Therefore, negative transformation is useful for **highlighting details present in dark regions of an image**.

2. LOG TRANSFORMATION

(a) Explain how log transformation enhances low-intensity pixels.

Problem Understanding

Log transformation is a grey level transformation technique used to enhance details in the **darker regions of an image**.

Transformation Formula

$$s = c \log(1 + r)$$

Where:

- **r** = Input pixel intensity
- **s** = Output pixel intensity
- **c** = Positive scaling constant

Method Selection

Log transformation is selected because it **expands low-intensity values and compresses high-intensity values**.

Therefore:

- **Dark regions become more visible.**
- **Low-intensity details are enhanced.**
- **High-intensity values are compressed.**
- **Details in darker areas can be identified more clearly.**

Solution Process

The steps involved are:

1. **Take the original pixel intensity r.**
2. **Add 1 to r.**
3. **Calculate $\log(1 + r)$.**
4. **Multiply the result by the scaling constant c.**
5. **Obtain the transformed intensity s.**

Formula

$$s = c \log(1 + r)$$

Interpretation of Results

Log transformation is particularly useful when important information is present in **low-intensity regions**.

It **enhances dark pixels**, expands the lower intensity range, and **compresses high-intensity values**.

Therefore, it improves the visibility of **details in darker regions**.

(b) Compute the log-transformed values.

Using:

$$s = c \log(1 + r)$$

Assume:

$$c = 1$$

Therefore:

$$s = \log(1 + r)$$

For r = 0

$$s = \log(1 + 0)$$

$$s = \log(1)$$

$$s = 0$$

Therefore:

$$0 \rightarrow 0$$

For r = 1

$$s = \log(1 + 1)$$

$$s = \log(2)$$

$$s = 0.693$$

Therefore:

$$1 \rightarrow 0.693$$

For r = 2

$$s = \log(1 + 2)$$

$$s = \log(3)$$

$$s = 1.099$$

Therefore:

$2 \rightarrow 1.099$

For $r = 3$

$s = \log(1 + 3)$

$s = \log(4)$

$s = 1.386$

Therefore:

$3 \rightarrow 1.386$

Log Transformation Table

Input Intensity (r) Log-Transformed Value

0	0
1	0.693
2	1.099
3	1.386

Accuracy of Calculation

The final calculated values are:

0, 0.693, 1.099, 1.386

All values are calculated using:

$s = \log(1 + r)$

Interpretation of Results

The transformed values **increase logarithmically**.

The transformation provides **greater enhancement to low-intensity regions** while compressing higher intensity values.

Thus, **log transformation makes details in dark regions more visible**.

3. HISTOGRAM PROCESSING

(a) Explain the role of histogram equalization in contrast enhancement.

Problem Understanding

An **image histogram** represents the frequency distribution of different intensity levels in an image.

When most pixels are concentrated in a small intensity range, the image may have **poor contrast**.

Histogram equalization improves contrast by redistributing the intensity values over a **wider available range**.

It is useful for:

- **Improving image contrast**
- **Making hidden details more visible**
- **Redistributing pixel intensities**
- **Improving the overall visual quality of an image**

Method Selection

For histogram equalization, the **Cumulative Distribution Function (CDF)** is calculated from the histogram.

Equalization Formula

$s(k) = \text{floor}((L - 1) \times \text{CDF}(k) / N)$

Where:

- **L = Number of grey levels**
- **CDF(k) = Cumulative distribution function**
- **N = Total number of pixels**
- **s(k) = Equalized intensity**

For a **3-bit image**:

L = 2^3

L = 8

Therefore:

L - 1 = 7

(b) Given a 3-bit image with histogram, compute the cumulative distribution and equalized intensity values.

Given Histogram

Intensity Frequency

0	2
1	2
2	4

Intensity Frequency

3 8

Step 1 – Calculate Total Number of Pixels

$$N = 2 + 2 + 4 + 8$$

$$N = 16$$

Therefore:

$$\text{Total number of pixels} = 16$$

Step 2 – Calculate Cumulative Frequency

Cumulative frequency is obtained by adding the frequencies successively.

Intensity Frequency Cumulative Frequency

0	2	2
1	2	4
2	4	8
3	8	16

Therefore:

$$\text{Cumulative frequencies} = 2, 4, 8, 16$$

Step 3 – Calculate Cumulative Distribution

Formula

$$\text{CDF}(k) = \text{Cumulative Frequency} / N$$

Since:

$$N = 16$$

For intensity 0:

$$\text{CDF}(0) = 2 / 16$$

$$\text{CDF}(0) = 0.125$$

For intensity 1:

$$\text{CDF}(1) = 4 / 16$$

$$\text{CDF}(1) = 0.250$$

For intensity 2:

$$\text{CDF}(2) = 8 / 16$$

$$\text{CDF}(2) = 0.500$$

For intensity 3:

$$\text{CDF}(3) = 16 / 16$$

$$\text{CDF}(3) = 1.000$$

Step 4 – Calculate Equalized Intensity Values

Formula

$$s(k) = \text{floor}(7 \times \text{CDF}(k))$$

For intensity 0:

$$s(0) = \text{floor}(7 \times 0.125)$$

$$s(0) = \text{floor}(0.875)$$

$$s(0) = 0$$

For intensity 1:

$$s(1) = \text{floor}(7 \times 0.250)$$

$$s(1) = \text{floor}(1.750)$$

$$s(1) = 1$$

For intensity 2:

$$s(2) = \text{floor}(7 \times 0.500)$$

$$s(2) = \text{floor}(3.500)$$

$$s(2) = 3$$

For intensity 3:

$$s(3) = \text{floor}(7 \times 1.000)$$

$$s(3) = 7$$

Final Histogram Equalization Table

Intensity	Frequency	Cumulative Frequency	CDF	Equalized Intensity
0	2	2	0.125	0

Intensity	Frequency	Cumulative Frequency	CDF	Equalized Intensity
1	2	4	0.250	1
2	4	8	0.500	3
3	8	16	1.000	7

Accuracy of Calculation

The final equalized intensity mapping is:

$0 \rightarrow 0$

$1 \rightarrow 1$

$2 \rightarrow 3$

$3 \rightarrow 7$

Therefore:

Equalized intensity values = 0, 1, 3, 7

Interpretation of Results

The original intensity levels **0, 1, 2, and 3** are redistributed to **0, 1, 3, and 7**.

This expands the intensity range from **0–3 to 0–7**.

Since **7 is the maximum intensity value of a 3-bit image**, the available intensity range is used more effectively.

Therefore, **histogram equalization improves image contrast and makes important image details more distinguishable**.

OVERALL CONCLUSION

Grey level transformation, log transformation, and histogram equalization are important image enhancement techniques used in **Computer Vision**.

Negative Transformation reverses the intensity values and can highlight details in dark regions.

Log Transformation enhances low-intensity regions and compresses high-intensity values.

Histogram Equalization redistributes intensity values and improves the overall contrast.

These image enhancement techniques **improve image quality** and make important regions easier to identify for further Computer Vision tasks such as **image segmentation, feature extraction, and object identification**.